

Touchy surface

Microsoft Surface is multi-touch capable and can recognise up to 52 simultaneous touch points

Price of touch

15-inch TS monitors featuring the projected capacitive technology are available in India for around Rs. 20,000. Resistive TS cost about half of that

placed on the screen is reduced to X and Y coordinates. The HP TouchSmart All-in-One PC uses this technology as does Microsoft's surface computing concept, the Touch Wall.

Advantages

There is no need to overlay any screens, just the bezel containing the IR transmitters and receivers are needed. So visibility is maximum among TS types. Since the circuitry can be completely sealed in the bezel, this technology offers the reliability even in tough environments. Most preferred TS technology for large screens.

Disadvantages

Since normal sunlight also contains IR rays, these screens do not perform optimally when placed in areas with abundant sunlight. The density of the IR diodes determines the sensitive areas of the screen and detectable size of the object. Some of the narrow spread or newer technologies are touched next.

Hybrids

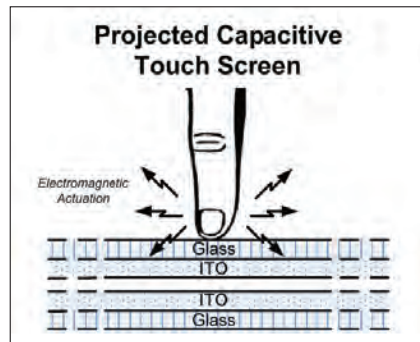
To get the best of both worlds - accuracy of tablets (not tablet PCs) and flexibility of human touch, some companies have come up with hybrid technology that combines the detection system used in tablets with capacitive TS technology. The Dell Latitude XT and HP TouchSmart TX2 are examples of devices using the hybrid technology created by a company called N-trig.

Wave transduction

This technology uses the property of glass to propagate sound or shock waves generated when touched. Receivers placed at 4 different sides of the screen can detect the waves, and by comparing the readings of each receiver the location of the point of touch on the screen is computed. This technology is fairly new, and also goes around under trademarks like Acoustic Pulse Recognition (audio waves are detected) and Dispersive Signal Technology (shock waves are detected).

Near field imaging

This is a proprietary technology, that relies on a proprietary layout of ITO film to detect touch events. The entire glass sheet is converted into a sensor,



Projected capacitive TS design

with its top side coated with ITO film in a particular layout, and the bottom layer entirely coated with ITO. A protective sheath is overlaid over the sensor. When activated, the sensor generates an electrostatic field, that is disrupted by touch with finger or grounded conducting object. The driver circuit continuously images the field, and change in the field is detected by comparison with previous images, and the point of touch is detected. NFI TS are considered the most rugged of TS and can withstand and work around dust and debris.



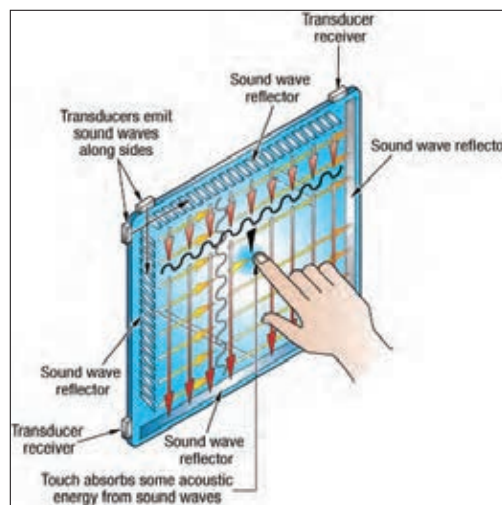
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Force detection

This technology uses sensors (called strain gauges) placed at the corners of the screen to detect the force applied to the screen during a touch operation. Based on the readings recorded at the four edges, the location of the point of touch is calculated.



Surface acoustic wave TS design

Optical technologies

This kind of TS relies on the principle of Frustrated Total Internal Reflection, and is utilised for transmitting light through fibre optic cables. Essentially, rather than have the IR source below the screen, it is placed around it, and the IR beams are projected through it at an angle such that the beams are internally reflected and don't escape the surface. When a finger touches the screen, at that point the internal reflection is frustrated, causing the IR rays to pass through the screen to the outside. This is picked up by the IR cameras placed behind/below the screen and used to detect touch and gestures.

Haptic technology

While not exactly a technology that is used to create touchscreens, Haptics is expected to play a major role in enhancing TS interfaces. Haptics, simply put, is similar to force feedback experienced in gaming devices. As TSs become popular and soft keypads take the place of normal keyboards, people used to touch typing will

be put off by the lack of proper sensory feedback from the flat screen. Haptics attempts to fill this void by creating the touch and feel of a normal key by simulating the edges and depressing action. This is just one application of haptics, which is already implemented in high end phones like the LG Cookie. Several phones offer Haptic experience, but this is mostly limited to a timely vibration as feedback for any action.

The future

Multi touch screens and collaborative software make a strong team. It is easy to picture a person using multi-touch to rotate an image, but the true power of multi touch is in having multiple people collaborating on the same project. Two people collaborating on the same screen would be better. But nothing would beat having a third person adding his inputs over the net. As internet connections speed up and appropriate software is created, such an intercontinental brainstorming session would be possible. TSs and gestures are predicted to make mice redundant. This is debatable.